Estimators and estimates

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Outline

Definition and properties of estimators

Confidence interval

Building estimators

Purpose of estimation

General case

- Find some **characteristics** of an **unknown** parameter regarding a **population** using only the **observations** from a sample of size n

In the following

- a **parameter** = a deterministic value (**not random**), fixed (and unknown)
- when a parameter is unknown, we will compute an **estimation** of the parameter based on the observations from a **sample**
  $\rightarrow$ the estimation depends on the sample
- **Caution:** an estimation can not give the true value of the parameter

Context

Given a parameter θ characterizing a population $\mathcal{P}$

- Let X be a random variable measuring a characteristics on individuals of $\mathcal{P}$ whose distribution depends on θ

- Let $(X_1, \ldots, X_n)$ be a n-sample[1] associated to X, and $(x_1, \ldots, x_n)$ some realizations (=observations) of $(X_1, \ldots, X_n)$

[1] i.e. $(X_1, \ldots, X_n)$ are independent and identically distributed (i.i.d.) random variables measured on a sample of size n with the same distribution as X, also noted $X_i \sim X$ or $X_i \sim \mathscr{L}(X)$

Estimator[1] and estimation[2]

Definition

- An **estimator**[1] of θ is a **random variable** noted $\widehat{\theta}(X)$ or $\widehat{\theta}(X_1, \ldots, X_n)$ built with $(X_1, \ldots, X_n)$ and a function $f : \mathbb{R}^n \to \mathbb{R}$, i.e.

$$\widehat{\theta}(X) = f(X_1, \ldots, X_n)$$

- An **estimate/estimation** of θ is a **realization/observation** of the corresponding estimator, it is noted $\widehat{\theta}(x)$ or $\widehat{\theta}(x_1, \ldots, x_n)$ and computed as

$$\widehat{\theta}(x) = f(x_1, \ldots, x_n)$$

[1] "*Estimateur*"
[2] "*Estimation*"

Estimator vs estimation: a summary

	type	notation	**short notation**[1]
Estimator	random variable	$\widehat{\theta}(X)$	$\widehat{\theta}$
Estimation	observation	$\widehat{\theta}(x)$	$\widehat{\theta}$

[1] **Caution:** no distinction between random variable and observation

Example

Context

- i.i.d. random variables on a sample of size n: $(X_1, \ldots, X_n)$
- corresponding observations on the sample: $(x_1, \ldots, x_n)$

Empirical mean/average[1]

- Estimator of the theoretical expectation[2] (random variable):

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$$

- Corresponding estimate/estimation (observation):

$$\bar{x}_n = \frac{1}{n} \sum_{i=1}^{n} x_i$$

[1] "*moyenne empirique*"

[2] "*espérance ou moyenne théorique*"

Estimator, estimation and uncertainty

- What **properties** for a **good** estimator?
- Estimation = **ponctual value** approximating the parameter
- With estimation and **sample variation**: objective to construct an interval containing the true value of the parameter with high probability

Properties of estimators: bias

Let $T_n = T_n(X_1, \ldots, X_n)$ be an estimator of a parameter θ

Estimator without bias[1] or asymptotically without bias[2]

- T_n is an estimator without bias if $\mathbb{E}(T_n) = \theta$
- T_n is an estimator asymptotically without bias if $\mathbb{E}(T_n) \xrightarrow[n \to \infty]{} \theta$

Interpretation

- Possible values for T_n are distributed around θ, and equal (or approximately equal for large n) to θ in average.

[1] "*estimateur sans biais*"

[2] "*asymptotiquement sans biais*"

Properties of estimators: convergence

Convergent estimator[1]

- T_n is a convergent estimator if $\mathrm{Var}(T_n) \xrightarrow[n\to\infty]{} 0$.
- T_n is an absolutely convergent estimator[2] if without bias and convergent.

Interpretation

- Possible values for T_n are more and more concentrated around θ as n grows.

Note: We can have two (different) absolutely convergent estimators T_n and T'_n for the same parameter θ. We will favour the one with the smallest variance[3].

[1] "*estimateur convergent ou correct*"
[2] "*estimateur absolument convergent*"
[3] It is being said to be more efficient ("*plus efficace*")

Empirical mean

Let X be a random variable with unknown mean μ and $(X_1 \ldots, X_n)$ an associated n-sample (i.e. i.i.d. and X_i follows the same distribution as X, also noted $X_i \sim X$ or $X_i \sim \mathscr{L}(X)$)

- **Empirical mean**: $\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$
- $\bar{X}_n$ is an estimator for μ **without bias and convergent** (i.e. absolutely convergent)

Why? **(exercise)**

Empirical mean

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Why? **(exercise)**

- $\mathbb{E}(\bar{X}_n) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}(X_i) = \frac{1}{n}\, n\, \mu = \mu$
- $\mathrm{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{1}{n^2}\, n\, \mathrm{Var}(X) = \frac{\mathrm{Var}(X)}{n} \xrightarrow[n \to \infty]{} 0$

Empirical frequency (proportion)

- Let p be an unknown proportion of *something* (e.g. a characteristic of individuals) in a population $\mathcal{P}$

- Rephrasing: let Y be a **Bernoulli variable** $\mathcal{B}(p)$ indicating if an individual of $\mathcal{P}$ is an occurrence of this *something*[1]. Let $(Y_1, \ldots, Y_n)$ be an associated n-sample[2]

[1] indicator $Y = 1$ if the considered individual is/has *something* and 0 otherwise

[2] $(Y_1, \ldots, Y_n)$ i.i.d. and indicator $Y_i \sim \mathcal{B}(p)$ such that $Y_i = 1$ if individual i is/has *something* and 0 otherwise

Empirical frequency (proportion)

Empirical frequency

- The empirical mean $\bar{Y}_n = \frac{1}{n}\sum_{i=1}^{n} Y_i$ corresponds to the **empirical frequency**¨:

$$\widehat{p} = \frac{\text{number of } \textit{something} \text{ in the sample}}{n}$$

- $\widehat{p} = \bar{Y}_n$ is an estimator for p **without bias and convergent** (i.e. absolutely convergent)

Why? **(exercise)**

Empirical frequency (proportion)

Empirical frequency

- The empirical mean $\bar{Y}_n = \frac{1}{n}\sum_{i=1}^{n} Y_i$ corresponds to the **empirical frequency**¨:

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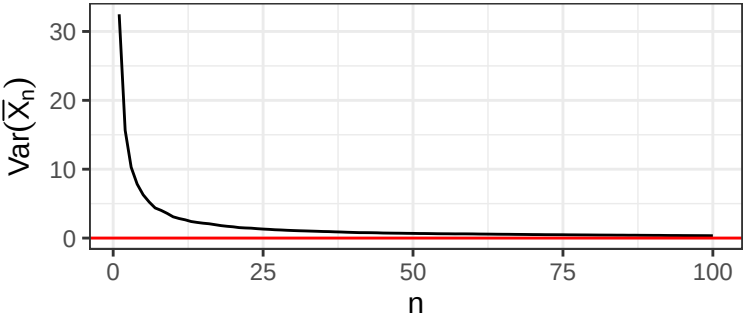
Why? **(exercise)**

- $\mathbb{E}(\bar{Y}_n) = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}(Y_i) = \frac{1}{n}\, n\, p = p$
- $\mathrm{Var}(\bar{Y}_n) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(Y_i) = \frac{1}{n^2}\, n\,\mathrm{Var}(Y_i) = \frac{p(1-p)}{n} \xrightarrow[n\to\infty]{} 0$

Illustration: variance of empirical mean

- $X \sim \mathcal{U}(-5, 15)$ and simulation of 1E4 realizations (observations) of $\bar{X}_n$ with $n = 1, \ldots, 100$
- Plot $n \mapsto \mathrm{Var}(\bar{X}_n)$



Empirical variance

Let X be a random variable with unknown mean μ and unknown variance σ^2, let $(X_1 \ldots, X_n)$ an associated n-sample (i.e. i.i.d. and $X_i \sim X$)

- **Intuition**[1] for the empirical variance:

$$S_*^2 = \frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X}_n)^2$$

- **Issue:** S_*^2 is biased (but asymptotically without bias and convergent) $\rightarrow$ exercise

- **Unbiased estimator** for the variance:

$$S^2 = \frac{n}{n-1} S_*^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X}_n)^2$$

[1] Reminder: $\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right]$

Proof (Exercise)

- $\mathbb{E}(S_*^2) = \frac{n-1}{n}\sigma^2 \neq \sigma^2$ but $\frac{n-1}{n}\sigma^2 \xrightarrow[n\to\infty]{} \sigma^2$
- $\mathrm{Var}(S_*^2) = \ldots \xrightarrow[n\to\infty]{} 0$

Hints:

$$S_*^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2 = \left(\frac{1}{n}\sum_{i=1}^{n}(X_i)^2\right) - (\bar{X}_n)^2$$

$$\mathbb{E}[(\bar{X}_n)^2] = \frac{1}{n^2}\left(\sum_{i=1}^{n}\mathbb{E}[(X_i)^2] + \sum_{\substack{i,j=1\\ i\neq j}}^{n}\mathbb{E}[X_iX_j]\right)$$

Outline

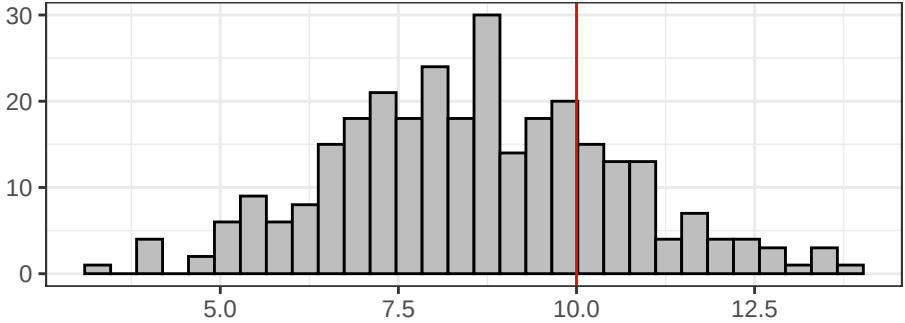
Definition and properties of estimators

Confidence interval

Building estimators

Ponctual estimation

Estimate = a single value (which depends on the sample)



- In grey/black: empirical distribution (histogram) of $\bar{X}_n$ (over 300 samples of size $n = 100$)
- In red (—): theoretical expectation value

Uncertainty

How to quantify the uncertainty of the sampling?

$\rightarrow$ **probability distribution** of the estimator

Using the probability distribution of the estimator

- we want to construct an **interval** that contains the parameter **true value** θ with high probability (i.e. close to 1)

Context

- Let X be a random variable whose distribution depends on a parameter θ and $(X_1, \ldots, X_n)$ an associated n-sample

- Let $\widehat{\theta} = T_n(X_1, \ldots, X_n)$ be an estimator (whose probability distribution is known) of a parameter θ

Variation interval vs confidence interval

- **Variation interval**[2]: the population parameter θ is **known** and we want an interval containing the **estimator** $\widehat{\theta}$ with high probability.

- **Confidence interval**: the population parameter θ is **unknown**, we want an interval containing the **true value** of θ with high probability. To do so, we use an estimator $\widehat{\theta}$ and its associated estimation.

[2] c.f. previous course

Confidence interval[1]

Definition

- Let T_n be an estimator for an unknown parameter θ
- A **confidence interval**[1] (of level $1-\alpha$) for the parameter θ, noted $[B_1\ ;\ B_2]$ or $B_1 \leq \theta \leq B_2$, is defined thanks to the probability distribution of the estimator T_n such that

$$\mathbb{P}(B_1 \leq \theta \leq B_2) = 1 - \alpha$$

  where **the boundaries**[2] B_1 and B_2 **depend on the estimator** T_n, i.e. $B_1 = u(T_n)$ and $B_2 = v(T_n)$ for two given functions $u : \mathbb{R} \to \mathbb{R}$ and $v : \mathbb{R} \to \mathbb{R}$.
- **Important**: The **boundaries** B_1 and B_2 are **random variables** (because defined with the random variable T_n).

[1] "*intervalle de confiance*"
[2] "*bornes*"

Comments on confidence interval

- The probability that the interval $[B_1\ ;\ B_2]$ contains the parameter value θ is $1-\alpha$.
- $1-\alpha$ is the **confidence level**[1].
- Generally we take $\alpha = 0.05$ (5%), so that $1-\alpha = 95\%$.

[1] "*niveau de confiance*"

Shape of a confidence interval

Unilateral or one-sided

Boundary on a single side

- right unilateral interval[1]

$$B_1 \leq \theta \quad \text{or} \quad [B_1\,;\,+\infty[$$

with $\mathbb{P}(B_1 \leq \theta) = 1 - \alpha$

- left unilateral interval[2]

$$\theta \leq B_2 \quad \text{or} \quad]-\infty\,;\,B_2]$$

with $\mathbb{P}(\theta \leq B_2) = 1 - \alpha$

[1] "*intervalle de confiance unilateral à droite ou inférieur*"
[2] "*intervalle de confiance unilateral à gauche ou supérieur*"

Shape of a confidence interval

Bilateral or two-sided

- **Bounded on both sides** $B_1 \leq \theta \leq B_2$

- Let $\alpha_1, \alpha_2 \in [0, 1]$ such that´ $\mathbb{P}(\theta < B_1) = \alpha_1$ and $\mathbb{P}(\theta \leq B_2) = 1 - \alpha_2$ then[3]

$$\mathbb{P}(B_1 \leq \theta \leq B_2) = 1 - \alpha$$

  with $\alpha = \alpha_1 + \alpha_2$

- Symmetric interval: $\alpha_1 = \alpha_2 = \alpha/2$
- Asymmetric interval: $\alpha_1 \neq \alpha_2$

[3] because $\mathbb{P}(B_1 \leq \theta \leq B_2) = \mathbb{P}(\theta \leq B_2) - \mathbb{P}(\theta < B_1)$

Realization of a confidence interval

Given

- an estimator T_N constructed with a n-sample $(X_1, \ldots, X_n)$
- an estimate t_n computed with the observations $(x_1, \ldots, x_n)$

The **realization of the confidence interval** is

$$b_1 \leq \theta \leq b_2 \quad \text{or} \quad [b_1 \ ; \ b_2]$$

where $b_1 = u(t_n)$ and $b_2 = v(t_n)$ are realizations of the random boundaries.

Important: the interval $b_1 \leq \theta \leq b_2$ is **not random**, the boundaries are fixed values, it makes no sense to say that $[b_1 \ ; \ b_2]$ contains the true value θ with probability 95%.

Confidence interval: in practice

- Let $Y = f(X_1, \ldots, X_n, \theta)$ be a random variable defined with $X_1, \ldots, X_n$, the parameter θ, and a function $f\colon \mathbb{R}^{n+1} \mapsto \mathbb{R}$ such that the **probability distribution** of Y is **known**.

- From the distribution of Y, we can find **quantiles** q_1 and q_2 such that

$$\mathbb{P}(q_1 \leq f(X_1, \ldots, X_n, \theta) \leq q_2) = 1 - \alpha$$

- By **resolving** $q_1 \leq f(X_1, \ldots, X_n, \theta) \leq q_2$ we can find an **interval** for θ.

Empirical mean

Let X be a random variable of mean μ and variance σ^2, let $(X_1, \ldots, X_n)$ be an associated n-sample

Reminder: the empirical mean is $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$

Two cases:

1. $X \sim \mathcal{N}(\mu, \sigma^2)$
2. n large and X admits an expectation μ and a variance $\sigma^2 < \infty$

Consequences:

$$\bar{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{and} \quad \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

because properties of Gaussian distribution (case 1) or central limit theorem (case 2)

Confidence interval for the empirical mean (with known[3] σ^2)

We have then

$$\mathbb{P}\left(-z_{1-\alpha/2} \leq \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq z_{1-\alpha/2}\right) = 1 - \alpha$$

where z_α is the quantile of level α for the standard Gaussian distribution.

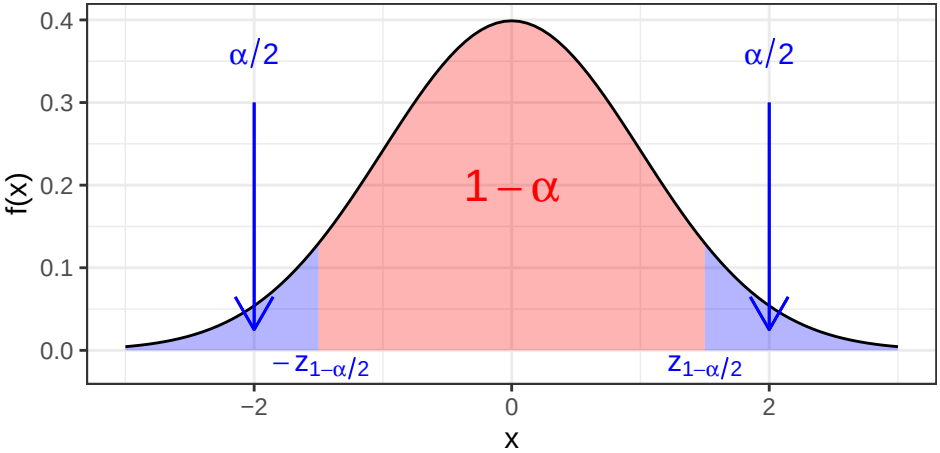
The **confidence interval** for μ when σ^2 is **known**:

$$\bar{X}_n - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X}_n + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}} \quad \text{i.e.} \quad \left[\bar{X}_n - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\ ;\ \bar{X}_n + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

The **realization of this confidence interval** is $\left[\bar{x}_n - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\ ;\ \bar{x}_n + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$

[3] Caution: it is generally not verified

Reminder: standard Gaussian distribution



Link between Gaussian, χ^2 and Student distributions

Context:

- $Z \sim \mathcal{N}(0,1)$
- $V \sim \chi^2(\ell)$ with $\ell \in \mathbb{N}$

Consequence:

$$T = \frac{Z}{\sqrt{\frac{V}{\ell}}} \sim \mathcal{T}(\ell)$$

"The ratio between a standard Gaussian distribution and the squared root of a renormalized χ^2 distribution with ℓ degrees of freedom is a Student's distribution with ℓ degrees of freedom."

Confidence interval for the empirical mean (with unknown σ^2)

If σ^2 is **unknown**: we replace σ by $S = \sqrt{S^2}$ where S^2 is the unbiased estimator of the variance:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$$

- We have $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1)$ (explained before)
- Thanks to *Cochran theorem*: $(n-1)\frac{S^2}{\sigma^2} \sim \chi^2(n-1)$

Consequence:

$$\frac{\bar{X}_n - \mu}{S/\sqrt{n}} \sim \mathcal{T}(n-1)$$

i.e. a Student's distribution with $n-1$ degrees of freedom[1]

[1] c.f. previous slide by using $Z = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$, $V = (n-1)\frac{S^2}{\sigma^2}$ and $\ell = n-1$

Confidence interval for the empirical mean (with unknown σ^2)

Note: the Student's density is symmetrical regarding axis $x = 0$.

We have then

$$\mathbb{P}\left(-t_{1-\alpha/2;n-1} \leq \frac{\bar{X}_n - \mu}{S/\sqrt{n}} \leq t_{1-\alpha/2;n-1}\right) = 1 - \alpha$$

where $t_{\alpha;n-1}$ is the quantile of level α for the Student's $\mathcal{T}(n-1)$ distribution.

Confidence interval for the empirical mean (with unknown σ^2)

The **confidence interval** for μ when σ^2 is **unknown** is

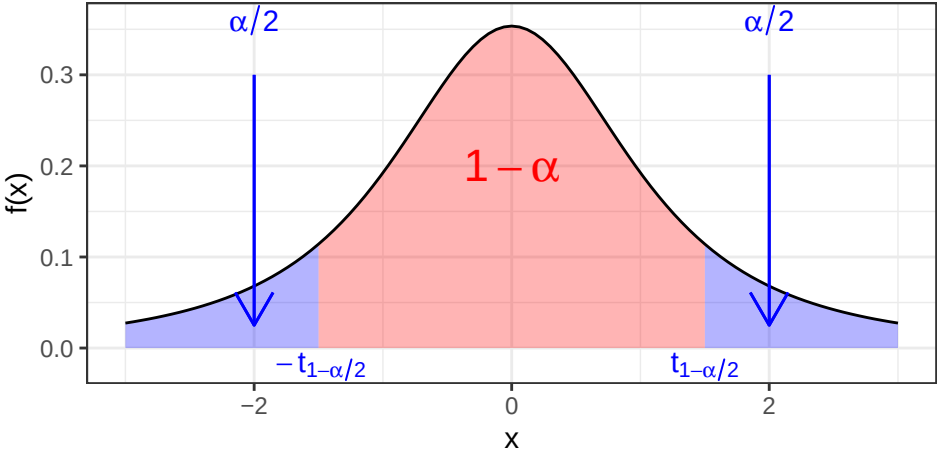
$$\bar{X}_n - t_{1-\alpha/2;n-1}\frac{S}{\sqrt{n}} \leq \mu \leq \bar{X}_n + t_{1-\alpha/2;n-1}\frac{S}{\sqrt{n}} \quad \text{i.e.} \quad \left[\bar{X}_n - t_{1-\alpha/2;n-1}\frac{S}{\sqrt{n}}\ ;\ \bar{X}_n + t_{1-\alpha/2;n-1}\right.$$

The **realization of this confidence interval** is

$$\left[\bar{x}_n - t_{1-\alpha/2;n-1}\frac{s}{\sqrt{n}}\ ;\ \bar{x}_n + t_{1-\alpha/2;n-1}\frac{s}{\sqrt{n}}\right]$$

where $s = \sqrt{s^2}$ and $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x}_n)^2$ is the realization of S^2, computed with $(x_1, \ldots, x_n)$ and $\bar{x}_n$

Illustration: Student's distribution



Important: $\mathcal{T}(n) \underset{n\to\infty}{\sim} \mathcal{N}(0,1)$

Application: confidence interval for a proportion

Context

- Let p be an unknown proportion of *something* in a population $\mathcal{P}$
- Rephrasing: $Y \sim \mathcal{B}(p)$ and $(Y_1 \ldots, Y_n)$ an associated n-sample, i.e. i.i.d. and $Y_i \sim \mathcal{B}(p)$

Reminder

- The empirical mean $\bar{Y}_n = \frac{1}{n}\sum_{i=1}^{n} Y_i$ corresponds to the empirical frequency

$$\widehat{p} = \frac{\text{number of } \textit{something} \text{ in the sample}}{n}$$

- $\widehat{p} = \bar{Y}_n$ is an estimator **without bias and convergent** for p

Application: confidence interval for a proportion

- Thanks to the central limit theorem:

$$\frac{\widehat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim \mathcal{N}(0,1)$$

Usage condition: $n > 100$, $np > 5$, $n(1-p) > 5$ (i.e. unknown value $5\% < p < 95\%$)

- **Consequence:**

$$\mathbb{P}\left(-z_{1-\alpha/2} \leq \frac{\widehat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \leq z_{1-\alpha/2}\right) = 1 - \alpha$$

where z_α is the quantile of level α for the standard Gaussian distribution.

Application: confidence interval for a proportion

- **Problem:** variance $\frac{p(1-p)}{n}$ depends on unknown p

- **Solution:** asymptotically[1] we can replace p by $\widehat{p}$ and preserve the Gaussian property, i.e.

$$\mathbb{P}\left(-z_{1-\alpha/2} \leq \frac{\widehat{p}-p}{\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}} \leq z_{1-\alpha/2}\right) \approx 1-\alpha$$

[1] when n is large

Application: confidence interval for a proportion

The **confidence interval** for p is then

$$\widehat{p} - z_{1-\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} \leq p \leq \widehat{p} + z_{1-\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}$$

where z_α is the quantile of level α for the standard Gaussian distribution.

The realization of this confidence interval is
$\widehat{p} - z_{1-\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} \leq p \leq \widehat{p} + z_{1-\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}$ **where** $\widehat{p} = \bar{y}_n$.

Caution: when computing the boundaries, you need to **check** *a posteriori* the validity of the assumptions made to use central limit theorem (i.e. $5\% < p < 95\%$)

Sample size and interval length

- The length of a confidence interval usually depends on the size n
  $\rightarrow$ the larger n, the narrower the interval.

- Example:

$$\bar{X}_n - t_{1-\alpha/2;n-1}\frac{S}{\sqrt{n}} \leq \mu \leq \bar{X}_n + t_{1-\alpha/2;n-1}\frac{S}{\sqrt{n}}$$

- Half interval length: $\Delta = t_{1-\alpha/2;n-1}\frac{S}{\sqrt{n}} \xrightarrow[n\to\infty]{} 0$

Consequence: we can choose n so that $|\bar{X}_n - \mu| < \Delta$ for a given Δ.

Reminder

- We generally note $\widehat{\theta}$ the estimator of a parameter θ Example: $\widehat{\mu}$, $\widehat{\sigma}$, $\widehat{\sigma}^2$, $\widehat{p}$
- **Advantage:** it simplifies the notations
- **Caution:** possible confusion between estimator and estimate/estimation (same notation)

Always keep in mind:

- Estimator = a random variable
- Estimate/estimation = a realization/an observation

Exercise 1

Before an election between two candidates (called A and B), a poll[1] is made to estimate voting intention. The polling organization publishes the following results:

- Candidate A: 52%
- Candidate B: 48%

(500 persons were questions for this poll).

Questions: What can you say about the expected result? Can the candidate A expect to win?

[1] "*sondage*"

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(500 persons were questions for this poll).

Questions: What can you say about the expected result? Can the candidate A expect to win?

Hint:

- use a confidence interval on the vote intention
- if $Z \sim \mathcal{N}(0,1)$ then $\mathbb{P}(-1.96 \leq Z \leq 1.96) = 0.95$

[1] "*sondage*"

Exercise 2

A company designs and builds pieces for planes (some screws[1]). They produce 10 000 pieces per day. The size of the screws is critical. The specification requires that the screw size is 25mm $\pm$ 1mm. The company sets up a quality control procedure to assess the conformity of its production regarding the specification. They sample 1000 pieces each day. On the first day, they find $\bar{x}_n = 25.5$ and $s^2 = 4$.

Question: would you assume the production to respect the specification?

[1] "*vis*"

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Question: would you assume the production to respect the specification?

Hint:

- $\mathcal{T}(n-1) \approx \mathcal{N}(0,1)$ for large n (e.g. $n > 500$)
- if $Z \sim \mathcal{N}(0,1)$ then $\mathbb{P}(-1.96 \leq Z \leq 1.96) = 0.95$

[1] "*vis*"

Solution 1

- We are going to compute a confidence interval on the proportion of vote for candidate A
- We have $\widehat{p} = 0.52$
- Reminder about confidence interval for a proportion at level $1 - \alpha$

$$\widehat{p} - z_{1-\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} \leq p \leq \widehat{p} + z_{1-\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}$$

- Realization of the confidence interval wih $1 - \alpha = 0.95$ (i.e. using $z_{1-\alpha/2} = 1.96$): $[0.476\ ;\ 0.564]$
- Comment: uncertainty about the vote result is larger than the difference between the voting intention for each candidate, thus this poll gives no information about the potential winner

Solution 1

- Realization of the confidence interval wih $1 - \alpha = 0.95$ (i.e. using $z_{1-\alpha/2} = 1.96$): [0.476 ; 0.564]
- Comment: the sampling uncertainty about the vote result is larger than the difference between the voting intention for each candidate, thus this poll gives no information about the potential winner.
- **Note:** if we do the same exercice with voting intention being 80%/20%, the realization of the confidence interval becomes [0.765 ; 0.835], the difference between the voting intention become larger than the sampling uncertainty and the poll becomes informative about the potential winner.

Solution 2

- In this case, we know that (large n)

$$\frac{\bar{X}_n - \mu}{S/\sqrt{n}} \sim \mathcal{T}(n-1)$$

- Here, $n > 500$ then $\mathcal{T}(n-1) \approx \mathcal{N}(0,1)$, then the confidence interval for μ when σ^2 is unknown is:

$$\bar{X}_n - z_{1-\alpha/2}\,\frac{S}{\sqrt{n}} \leq \mu \leq \bar{X}_n + z_{1-\alpha/2}\,\frac{S}{\sqrt{n}}$$

  where z_α is the quantile of level α for the standard Gaussian distribution.

- Realization of the interval at level $1-\alpha = 0.95$ with $n = 1000$, $\bar{x}_n = 25.5$ and $s^2 = 4$:

$$[25.38\,;\,25.62]$$

Outline

Definition and properties of estimators

Confidence interval

Building estimators

Notations

- $\mathcal{P}$ population
- X random variable
- $\theta \in \Theta$ unknown parameter ($\Theta =$ set of possible values for θ)
- if X is **continuous**: $x \mapsto f_\theta(x)$ density of X
- if X is **discrete**: $x \mapsto P_\theta(x)$ probability function of X i.e. $P_\theta(x) = \mathbb{P}(X = x)$
- $(X_1, \ldots, X_n)$ random (i.i.d.) sample of size n associated with X
- $(x_1, \ldots, x_n)$ observations in the sample

Likelihood[1]

Likelihood[1] of an iid random sample

- if X is **discrete**

$$\mathcal{L}(x_1, \ldots, x_n; \theta) = \mathbb{P}(X_1 = x_1, \ldots X_n = x_n) = \prod_{i=1}^{n} P_\theta(x_i)$$

- if X is **continuous**

$$\mathcal{L}(x_1, \ldots, x_n; \theta) = \prod_{i=1}^{n} f_\theta(x_i)$$

$\rightarrow$ density of the multivariate probability distribution of the random vector $(X_1, \ldots, X_n)$

[1] "*Vraisemblance*"

Log-likelihood[1]

Log-likelihood[1] of a random sample

- if X is **discrete**

$$\log \mathcal{L}(x_1, \ldots, x_n; \theta) = \sum_{i=1}^{n} \log P_\theta(x_i)$$

- if X is **continuous**

$$\log \mathcal{L}(x_1, \ldots, x_n; \theta) = \sum_{i=1}^{n} \log f_\theta(x_i)$$

We generally note $\ell(x_1, \ldots, x_n; \theta) = \log \mathcal{L}(x_1, \ldots, x_n; \theta)$

[1] "*Log-vraisemblance*"

Why taking the log?

Example: $X \sim \mathcal{N}(\mu, \sigma^2)$

$$\mathcal{L}(x_1, \ldots, x_n; \mu, \sigma) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_i-\mu}{\sigma}\right)^2}$$

$$\ell(x_1, \ldots, x_n; \mu, \sigma) = -n\,\log(\sigma) - \frac{n}{2}\log(2\pi) - \frac{1}{2\,\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

Statistical analysis

1. Question about an unknown parameter θ on a population $\mathcal{P}$

2. **Model:**
   - Define a random variable X measuring a quantity on individuals of $\mathcal{P}$
   - Choose a probability distribution for X, parametrized by θ, based on theoretical results (e.g. central limit theorem), or that is credible/plausible/likely regarding the characteristics of the population $\mathcal{P}$

3. **Analysis:**
   - Write the likelihood associated to the model
   - Find the **most probable** value $\widehat{\theta}$ for θ knowing the data $(x_1, \ldots, x_n)$, i.e. **maximize** the likelihood $\mathcal{L}(x_1, \ldots, x_n; \theta)$ or log-likelihood $\ell(x_1, \ldots, x_n; \theta)$ regarding θ

$\rightarrow$ **optimization**

Maximum likelihood estimator (MLE)[1]

The **maximum likelihood estimator** $\widehat{\theta}_{ML}$ is defined as[2]

$$\widehat{\theta}_{ML} = \underset{\theta \in \Theta}{\operatorname{argmax}}\ \mathcal{L}(X_1, \ldots, X_n; \theta) = \underset{\theta \in \Theta}{\operatorname{argmax}}\ \ell(X_1, \ldots, X_n; \theta)$$

It verifies

$$\mathcal{L}(X_1, \ldots, X_n; \widehat{\theta}_{ML}) = \max_{\theta \in \Theta} \mathcal{L}(X_1, \ldots, X_n; \theta) = \max_{\theta \in \Theta} \ell(X_1, \ldots, X_n; \theta)$$

[1] "*Estimateur du maximum de vraisemblance (EMV)*"

[2] argmax = "argument (value) that maximize"

Maximum likelihood estimate (MLE)

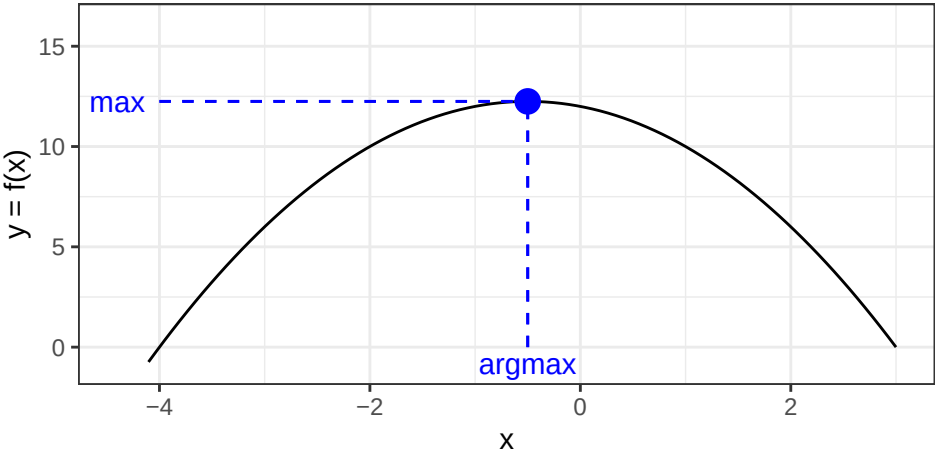
The **maximum likelihood estimate** $\widehat{\theta}_{ML}$ is the corresponding realization

$$\widehat{\theta}_{ML} = \underset{\theta \in \Theta}{\operatorname{argmax}}\ \mathcal{L}(x_1, \ldots, x_n; \theta) = \underset{\theta \in \Theta}{\operatorname{argmax}}\ \ell(x_1, \ldots, x_n; \theta)$$

It verifies

$$\mathcal{L}(x_1, \ldots, x_n; \widehat{\theta}_{ML}) = \max_{\theta \in \Theta}\ \mathcal{L}(x_1, \ldots, x_n; \theta) = \max_{\theta \in \Theta}\ \ell(x_1, \ldots, x_n; \theta)$$

Illustration: max and argmax



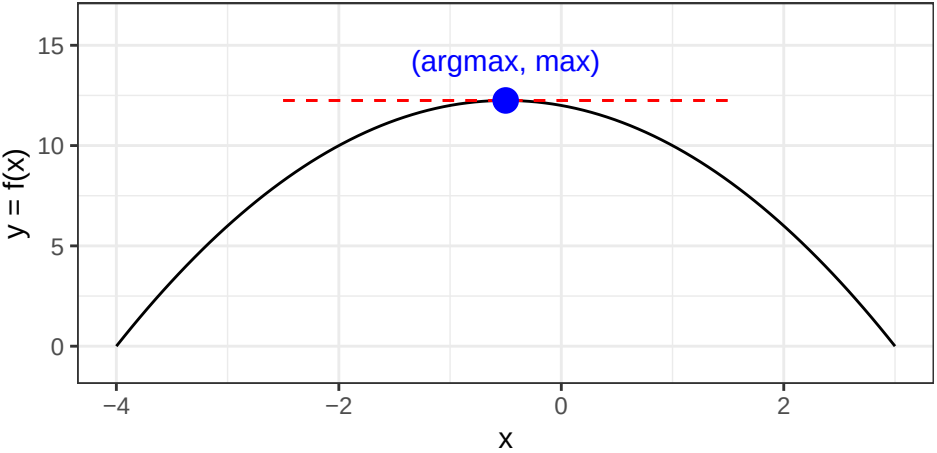
Optimization and likelihood regularity

- To find the MLE value, we need to find the max and argmax of the likelihood.
- **Conditions:** convexity and regularity of the objective function (!! may not be verified !!)
- Under regularity assumptions, the derivative of the likelihood (or log-likelihood) is null when $\theta = \widehat{\theta}_{ML}$ i.e.

$$\left.\frac{\partial \mathcal{L}(x_1, \ldots, x_n; \theta)}{\partial \theta}\right|_{\theta=\widehat{\theta}_{ML}} = 0$$

- **Solution:** solve the previous equality to find $\widehat{\theta}_{ML}$

Illustration: derivative is null

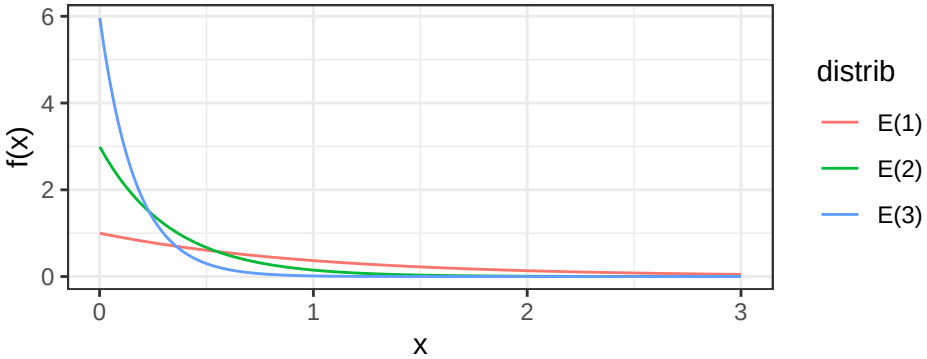


Tangent slope = 0

Example

Assume that $X \sim \mathcal{E}(\lambda)$ i.e. exponential distribution of parameter λ

Density: $f_\lambda(x) = \mathbf{1}_{\{x \geq 0\}} \lambda \, e^{-\lambda x}$



Example

Likelihood: $\mathcal{L}(x_1, \ldots, x_n; \lambda) = \prod_{i=1}^{n} \lambda\, e^{-\lambda\, x_i} = \lambda^n \prod_{i=1}^{n} e^{-\lambda\, x_i}$ (assuming all $x_i \geq 0$)

Log-likelihood: $\ell(x_1, \ldots, x_n; \lambda) = n \log(\lambda) + \sum_{i=1}^{n} -\lambda\, x_i$

Compute the derivative:

$$\frac{\partial}{\partial \lambda} \ell(x_1, \ldots, x_n; \lambda) = \frac{n}{\lambda} - \sum_{i=1}^{n} x_i$$

Find the optimum i.e. the value $\widehat{\lambda}_{ML}$ that set the derivative equal to 0

$$\frac{n}{\widehat{\lambda}_{ML}} - \sum_{i=1}^{n} x_i = 0 \qquad \Rightarrow \qquad \widehat{\lambda}_{ML} = \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\bar{x}_n}$$

Note:´ $\mathbb{E}(X) = \frac{1}{\lambda}$

Exercise

Assume that $X \sim \mathcal{N}(\mu, \sigma^2)$

Find the maximum likelihood estimate for μ and σ^2

Solution

- $\widehat{\mu}_{ML} = \frac{1}{n}\sum_{i=1}^{n} x_i$
- $\widehat{\sigma}^2_{ML} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x}_n)^2$

Log-likelihood:

$$\ell(x_1, \ldots, x_n; \mu, \sigma) = -n\,\log(\sigma) - \frac{n}{2}\log(2\pi) - \frac{1}{2\,\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

Partial derivative regarding μ:

$$\frac{\partial}{\partial\,\mu}\ell(x_1, \ldots, x_n; \mu, \sigma) = -\frac{1}{2\,\sigma^2}\sum_{i=1}^{n} -2(x_i - \mu)$$

Partial derivative regarding σ:

$$\frac{\partial}{\partial\,\sigma}\ell(x_1, \ldots, x_n; \mu, \sigma) = -\frac{n}{\sigma} + \frac{1}{\sigma^3}\sum_{i=1}^{n}(x_i - \mu)^2$$

Note on likelihood optimization

- θ can be a **set** of parameter (e.g. $\theta = (\mu, \sigma)$ in the previous example)
- Here we assumed enough **regularity** on the log-likelihood so that we can use the values of parameters so that the corresponding partial derivatives are null to find the global optimum
- **Not always true** $\rightarrow$ convex optimization (e.g. gradient descent, or non convex optimization[4])

[4] more complicated

Properties of the maximum likelihood estimator

- If an estimator without bias and with bounded[1] variance exists, it is equal to the MLE (Cramér-Rao)
- Cramér-Rao bound $= \frac{1}{I_{n,\theta}}$ where $I_{n,\theta} = -\mathbb{E}\left(\frac{\partial^2}{\partial\theta^2}\ell(X_1,\ldots,X_n;\theta)\right)$ is the Fisher information
- For large n, the MLE becomes unique and is convergent, i.e. converges in probability toward θ

$$\widehat{\theta}_{ML} \xrightarrow[n\to\infty]{p} \theta$$

- The MLE can be biased (example of $\widehat{\sigma}_{ML}$ for Gaussian distribution)
- The MLE is asymptotically Gaussian $\widehat{\theta}_{ML} \underset{n\to\infty}{\sim} \mathcal{N}(\theta, 1/I_{n,\theta})$

[1] "*borné*"

Estimation with the method of moments[1]

- $\bar{X}_n^r = \frac{1}{n}\sum_{i=1}^n X_i^r$ are the empirical moments of order r
- $\mu_r = \mathbb{E}(X^r)$ are the theoretical moments of order r

(When $r = 1$, it corresponds to empirical and theoretical means respectively)

The estimator $\widehat{\mu}_r$ by the **method of moments** is $\bar{X}_n^r$.

Example: MLE for the parameter of an exponential distribution, or MLE for mean and variance of Gaussian distribution (c.f. previously)

[1] "*Méthode des moments*"

Properties of the estimator by the method of moments

- It gives convergent estimators
- Using the method of moments is more intuitive than building the MLE (and generally more simple)
- The method of moments gives imprecise estimators when n is not large
- Problem: in general asymptotic distribution is unknown, less properties than MLE

Questions?